

Title : Project Registration & Progress Review**FF No. 180**

Department: AIML	Academic Year: 2025-26
Semester : 3	Group No. : 5
Project Title: AI and Blockchain-Based System for Visual Quality Grading and Traceability of Fruits in Supply Chains	
Project Area: Agricultural Technology (AgriTech) & Blockchain	

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Project approved / Not approved		
Guide	Project Coordinator	Head of Department

Project Synopsis

Project Title:

AI and Blockchain-Based System for Visual Quality Grading and Traceability of Fruits in Supply Chains

Problem Statement:

In traditional agricultural supply chains, there is a lack of reliable, transparent, and tamper-proof systems for evaluating and tracking the quality of produce like fruits. Manual fruit grading is subjective, error-prone, and lacks traceability — making it difficult for stakeholders to verify the quality and origin of agricultural goods.

Abstract:

This project aims to automate the fruit grading process using a machine learning model and secure the grading results using blockchain technology. The ML model will classify fruits into grades (A, B, C) based on visual features. The grading information, along with metadata (date, location, quality parameters), will then be stored on a decentralized blockchain ledger to ensure immutability and transparency. This system is especially beneficial for stakeholders across the agricultural supply chain, ensuring fair trade, reducing fraud, and improving product quality tracking.

Introduction:

Agricultural product grading is vital for determining pricing, storage, and distribution decisions. However, current grading systems often rely on manual inspections that vary based on human judgment and are not reliably documented. By combining machine learning with blockchain, this project introduces a smart, traceable, and transparent fruit grading system that enhances both quality assurance and supply chain credibility.

Objectives:

- To automate the fruit grading process using a trained machine learning model.
- To ensure transparency and traceability of grading results using blockchain.
- To eliminate tampering or misreporting of product quality.
- To provide stakeholders with easy access to fruit quality history via decentralized records.
- To improve efficiency and trust across the agricultural supply chain.

Key Features:

- Real-time fruit image capture and grading.
- ML-based classification into A, B, or C grades.
- Secure storage of grading results on blockchain.
- Traceable grading history and metadata (location, timestamp).
- Web interface for viewing and verifying results.

- Tamper-proof and verifiable certification.

Technology Stack:

- **Model Training & Image Processing:** Python, TensorFlow, OpenCV
- **Backend & API Integration:** Flask, Supabase, QR Code Library
- **Blockchain & Decentralized Storage:** IPFS, Solidity, Polygon Testnet, MetaMask
- **Frontend Development:** HTML, CSS, JavaScript, Web3.js
- **Connectivity & Integration Tools:** MetaMask, Web3.js, QR Code Library

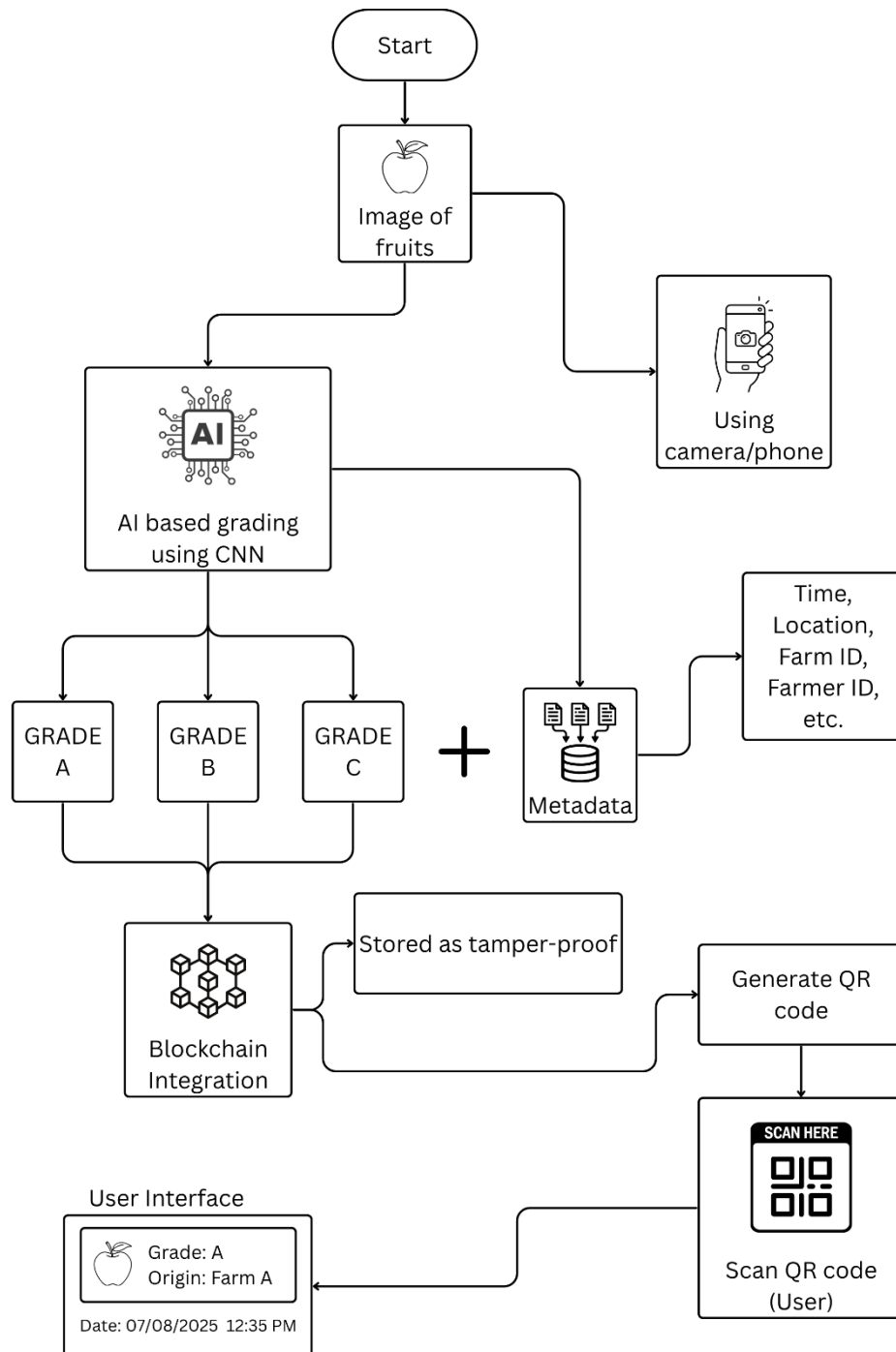
Benefits:

- Fully automated and consistent fruit quality evaluation.
- Immutable and tamper-proof grading records.
- Improved farmer-to-market trust and transparency.
- Fairer pricing and reduced product rejection rates.
- Easier audits, certifications, and compliance reporting.

Conclusion:

This project bridges the gap between agricultural quality control and modern-day digital trust systems. By integrating machine learning and blockchain, it provides a scalable, transparent, and intelligent fruit grading solution. The system empowers all stakeholders — from farmers to consumers — with reliable data, reducing fraud and inefficiencies in the food supply chain.

Workflow

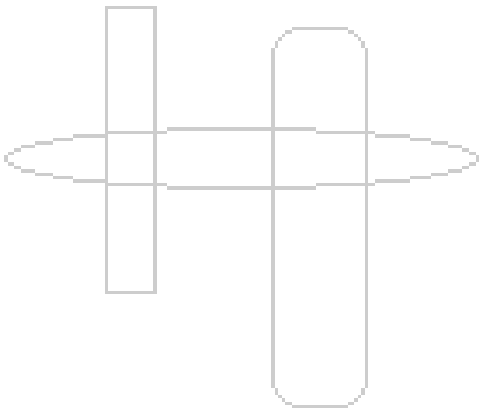


Literature Review

No	Title (Source, Date)	Project Idea	Tech Stack	Limitations	Our Improvement
1	“Deep learning-based fruit grading using hyperspectral and RGB imaging” (IEEE Trans. on Instrumentation, Aug 2024)	Use hyperspectral imaging + CNN to grade fruit ripeness and defects	CNN, hyperspectral + RGB sensors	Specialized expensive hardware, limited scalability	Use standard RGB, reduce cost, apply CNN/ViT and simpler capture
2	“Blockchain-based traceability for fresh produce supply chain” (IEEE Access, Sep 2024)	Record supply-chain events (farm, transport, storage) into blockchain	Ethereum smart contracts, off-chain DB, IoT sensors	No automated grading; manual data entry	Integrate automated ML grading as certified data input
3	“AI-enabled apple defect detection using vision transformers” (IEEE CCIS, Jul 2025)	ViT model to detect blemishes and grade apples	Vision Transformer (ViT), Python, OpenCV	Focus only on apples; no traceability	Extend to multiple fruit kinds and add blockchain storage
4	“Smart agriculture traceability using Hyperledger and ML forecasting” (IEEE Access, Nov 2024)	Use Hyperledger Fabric + ML to forecast demand and trace products	Hyperledger, Python ML, IoT	No image-based quality inspection	Add image-based grading pipeline feeding into the ledger
5	“ML-driven defect detection in citrus fruits via CNN ensemble” (IEEE Access, Jun 2025)	Ensemble CNN models classify citrus from class A-C	Multiple CNNs, large image dataset	Only citrus fruits; no metadata traceability	Broaden fruit set, integrate blockchain metadata storage

Group No.			
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

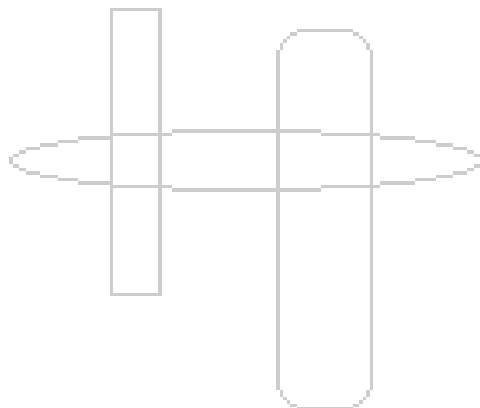
Review No.: 1	Group No.:	Date:
Progress Review Report		
		
Signature of Guide:		

Review No.: 2

Group No.:

Date:

Progress Review Report



Signature of Guide: